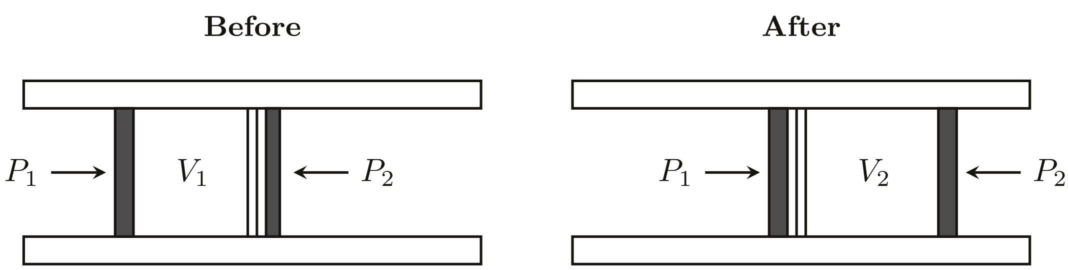
The Joule-Thomson Process

Source: D. Tong, Statistical Physics, Problem Sheet 4, Question 4.

This question describes the Joule-Thomson process (also known as the Joule-Kelvin process). The figure shows a thermally insulated pipe which has a porous barrier separating two halves of the pipe. A gas of volume V_1, initially on the left-hand side of the pipe, is forced by a piston to go through the porous barrier using a constant pressure P_1. Assume the process can be treated quasistatically. As a result the gas flows to the right-hand side, resisted by another piston which applies a constant pressure P_2 ($P_2 < P_1$). Eventually all of the gas occupies a volume V_2 on the right-hand side.



Joule-Thomson process: gas is pushed through a porous barrier from pressure P_1 to $P_2 < P_1$.

1. Show that enthalpy, $H = E + PV$, is conserved.
2. Find the Joule-Thomson coefficient $\mu_{\mathrm{JT}} \equiv \left(\frac{\partial T}{\partial P}\right)_H$ in terms of T, V, the heat capacity at constant pressure C_P, and the volume coefficient of expansion $\alpha \equiv \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_P$. (Hint: You will need to use a Maxwell relation.)
3. What is μ_{JT} for an ideal gas?
4. If we wish to use the Joule-Thomson process to cool a real (non-ideal) gas, what must the sign of μ_{JT} be?
5. Derive μ_{JT} for a gas obeying the van der Waals equation of state to leading order in the density N/V. For what values of temperature T can the gas be cooled?

Solution. Label all quantities on the left by 1 and quantities on the right by 2. We have

$$dH = dH_1 + dH_2 = dE_1 + P_1 dV_1 + V_1 dP_1 + dE_2 + P_2 dV_2 + V_2 dP_2. \tag{1.1}$$

Since the process is adiabatic, we have $dE_i = -\delta W_i = -P_i dV_i$. Meanwhile, the pressure is constant; thus, all quantities on the left vanish. Hence, we have $dH = 0$.

Now, consider the temperature T as a function of H and P. The total derivative of T is given by

$$dT = \left(\frac{\partial T}{\partial H}\right)_P dH + \left(\frac{\partial T}{\partial P}\right)_H dP = \left(\frac{\partial T}{\partial P}\right)_H dP \equiv \eta_{\mathrm{JT}} dP. \tag{1.2}$$

Using Maxwell relations, we have

$$\eta_{\mathrm{JT}} = -\left(\frac{\partial T}{\partial H}\right)_P \left(\frac{\partial H}{\partial P}\right)_T = -\frac{1}{C_P}\left[T\left(\frac{\partial S}{\partial P}\right)_T + V\right] = \frac{1}{C_P}\left[T\left(\frac{\partial V}{\partial T}\right)_P - V\right]. \tag{1.3}$$

For ideal gas,

$$\left(\frac{\partial V}{\partial T}\right)_P = V \implies \eta_{\mathrm{JT}} = 0. \tag{1.4}$$

Since $dP < 0$, for a cooling process, $dT < 0$, we require $\eta_{\mathrm{JT}} > 0$.

The van der Waals equation of state is

$$\left(P + \frac{aN^2}{V^2}\right)(V - Nb) = NkT. \tag{1.5}$$

Expanding and dropping the ab term as second order in the density N/V,

$$PV - PNb + \frac{aN^2}{V} = NkT. \tag{1.6}$$

Differentiating implicitly at constant P,

$$P\left(\frac{\partial V}{\partial T}\right)_P - \frac{aN^2}{V^2}\left(\frac{\partial V}{\partial T}\right)_P = Nk, \tag{1.7}$$

and so

$$\left(\frac{\partial V}{\partial T}\right)_P = \frac{Nk}{P - aN^2/V^2} \approx \frac{Nk}{P}\left(1 + \frac{aN^2}{PV^2}\right). \tag{1.8}$$

To leading order in N/V we may replace $V \approx NkT/P$ in the correction term only, giving

$$\left(\frac{\partial V}{\partial T}\right)_P \approx \frac{Nk}{P} + \frac{aN^2}{kT^2V}. \tag{1.9}$$

Substituting into the expression for η_{JT} and using $V \approx NkT/P + Nb - aN/kT$ to the same order,

$$T\left(\frac{\partial V}{\partial T}\right)_P - V = \frac{NkT}{P} + \frac{aN}{kT} - \frac{NkT}{P} - Nb + \frac{aN}{kT} = \frac{2aN}{kT} - Nb. \tag{1.10}$$

Hence,

$$\eta_{\mathrm{JT}} = \frac{N}{C_P}\left(\frac{2a}{kT} - b\right). \tag{1.11}$$

The gas can be cooled ($\eta_{\mathrm{JT}} > 0$) only below the *inversion temperature*,

$$T < T_{\mathrm{inv}} \equiv \frac{2a}{kb}. \tag{1.12}$$